

Experiment 6

Module Name: Robot Frameworks

Experiment Name: Tags and Test Execution

Problem Statement: Assign tags to test cases and test suites and execute only those tests with a specific tag using the "--include" or "--exclude" command-line options. Run your tests in parallel by utilizing the "--processes" option

Objective: Run your tests in parallel by utilizing the "--processes" option. To logically group test cases for regression or smoke testing and utilize multi-processing tools to reduce execution time.

Tools, Software, and Concepts Used:

Test Tagging ([Tags]), Command Line Interface (CLI) execution parameters, pabot (Parallel executor for Robot Framework)

Source Code / Implementation Steps:

1. Create a Folder inside that folder create 'Experiment6.robot' and write this:

Step 1: Assigning Tags in the .robot file

Code snippet

```
*** Settings ***
Library      SeleniumLibrary

*** Test Cases ***
Test Checkout Flow
    [Tags]    Smoke    Regression
    Log       Executing critical checkout flow...

Test Profile Picture Upload
    [Tags]    Regression
    Log       Executing non-critical profile upload...
```

Step 2: Now from the folder go to cmd and write this ~~Execution via CLI~~ To run only the "Smoke" tests:

>>> python -m robot --include Smoke Experiment6.robot

>>> python -m pabot.pabot --processes 2 Experiment6.robot

Output with clearly labelled screenshots:



Test Suite	Total	Passed	Failed	Skipped	Elapsed	Pass/Fail/Skip
Test Suite	1	2	0	0	00:00:00	2/2/0
Regression Suite	1	2	0	0	00:00:00	2/2/0
Experiment6	1	2	0	0	00:00:00	2/2/0

Screenshot 1: Output of log.html

```
D:\My Desktop\Robot Project>python -m robot --include Smoke Experiment6.robot
=====
Experiment6
=====
Test Checkout Flow                                     | PASS |
=====
Experiment6                                           | PASS |
=====
1 test, 1 passed, 0 failed
=====
Output:  D:\My Desktop\Robot Project\output.xml
Log:      D:\My Desktop\Robot Project\log.html
Report:   D:\My Desktop\Robot Project\report.html

D:\My Desktop\Robot Project>python -m pabot.pabot --processes 2 Experiment6.robot
Initialized logging in .\pabot_results
Robot Framework remote server at 127.0.0.1:8270 started.
Storing .pabotsuitenames file
2026-09-21 00:49:04.301668 [PID:12200] [0] [ID:0] EXECUTING Experiment6
2026-09-21 00:49:05.637541 [PID:12200] [0] [ID:0] PASSED Experiment6 in 1.3 seconds
2 tests, 2 passed, 0 failed, 0 skipped.
=====
Output:  D:\My Desktop\Robot Project\output.xml
Log:      D:\My Desktop\Robot Project\log.html
Report:   D:\My Desktop\Robot Project\report.html

Finalizing Pabot execution...
Stopping PabotLib process
Robot Framework remote server at 127.0.0.1:8270 stopped.
PabotLib process stopped
Total testing: 1.30 seconds
Elapsed time: 2.64 seconds
Logs flushed successfully.
```

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Screenshot 2: Terminal Output

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Brief Result: The CLI commands successfully filtered the test suite, executing only the targeted tests, and pabot successfully instantiated multiple execution threads.

Observation: Tags can be applied dynamically and act as metadata, allowing complex CI/CD pipeline routing without changing the code itself.

Conclusion: Tagging combined with parallel execution is essential for maintaining fast feedback loops in agile deployment environments.

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